PROJECT PROPOSAL | RESEARCH-BASED ENGINEERING PROJECT | INTELLIGENT SYSTEMS DESIGN LAB

Driver Drowsiness Detection System for Accident Prevention

A low-cost, real-time computer-vision prototype using facial landmarks, EAR / MAR and temporal analysis

CATEGORY
AI / Computer Vision / Road Safety

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1 Executive Summary

Drowsy driving slows reaction, narrows attention and can cause microsleeps that end in a lane departure or collision. India recorded 172,890 road-crash deaths in 2023 [1], but the national summary does not separate drowsiness, so its Indian share is unknown. MoRTH's current annual reporting does not publish a separate drowsiness category; related factors are collected in eDAR under "Long Distance covered / Driver Restless" [9]. This proposal describes a low-cost, non-contact prototype: a webcam feeds MediaPipe facial landmarks into Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR) and PERCLOS calculations, and temporal logic triggers an audible, optionally vibrating, alert. Performance will be measured experimentally; no accuracy is claimed in advance.

2 Problem Statement

Fatigue impairs alertness through slower reactions, attention lapses and microsleeps. India recorded 480,583 road crashes, 172,890 deaths and 462,825 injuries in 2023 [1]. Current national annual reporting does not publish a separate drowsiness share; a 2025 MoRTH parliamentary answer notes that revised reporting formats from 2017 did not include driver fatigue or drowsiness, while related factors are collected in eDAR under "Long Distance covered / Driver Restless" [9].

Data limitation. India's national road-accident totals are not drowsiness-specific. The current annual MoRTH table does not provide a separate national drowsiness category [1][9].

480,583
road crashes in India in 2023. All causes, not drowsiness-specific.
MoRTH, Road Accidents in India 2023 [1]

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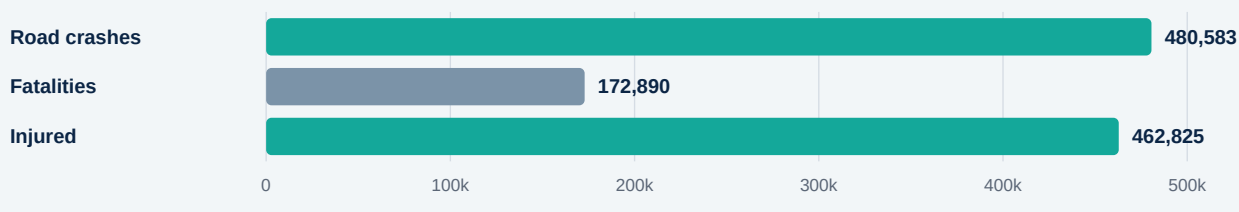
MoRTH, Road Accidents in India 2023 [1]

3 Research Evidence

India's 2023 national totals show the scale of road harm, but they do not provide a separate drowsiness percentage. The chart below therefore uses the official all-cause India figures rather than substituting a non-comparable foreign drowsiness estimate [1][9].



Source: MoRTH, Road Accidents in India 2023 [1]. All-cause totals; drowsiness is not separately reported.

Figure 1. Official India road-accident totals for 2023; the chart does not imply that all crashes were drowsiness-related.

Scope of the proposed system. It monitors *observable* signs of fatigue: prolonged eye closure, abnormal blinking, frequent yawning and, optionally, head pose. It cannot detect every form of fatigue (for example, cognitive fatigue with open eyes), and camera view, lighting, glasses and face orientation will affect it. Testing will use recorded footage or controlled settings, never drowsy driving on public roads.

4 Existing Technology Comparison

Approach	Method and hardware	Published result (dataset context)	Cost*	Real-time	Comfort / lighting	Privacy	Main limitations	How this project differs
Camera-based eye and face monitoring	EAR, MAR, head pose (or CNN) on facial video; RGB/IR	Up to 99% accuracy (Random Forest; 23 of 36 NTHU-DDD subjects, 70/30 split) [5]	Low (webcam)	Yes: 2–4 ms/frame on i7 [5]	Contactless; sensitive to lighting, glasses	Face video; needs consent, local processing	Face detector failures with several faces, changing light or dark roads [5]	**Lightweight MediaPipe with EAR/MAR/PERCLOS; persistence timers; on-device, low-cost**
EEG monitoring	Brain-signal features (e.g. alpha activity) with ML or CNN	78.35% mean accuracy, leave-one-subject-out, 11 subjects; baselines 53.4–72.7% [6]	Medium to high	Windowed; window sets latency	Electrodes on scalp, intrusive; unaffected by light	Neural data; highly sensitive	Large inter-subject variation makes calibration-free use hard [6]; motion artefacts	**No scalp electrodes; contactless camera sensing with simpler embedded setup**
Steering-wheel behaviour	Slow drift and corrective steering patterns	86.1% recognition, slight vs strong fatigue; simulator sleep-deprivation study, N = 12 [7]	Low if sensor exists	Yes; uses vehicle signals	Contactless; unaffected by light; affected by road	Low (vehicle data)	Not specific to drowsiness; road and driver dependent; validated mainly in simulators	**Directly observes facial fatigue cues; no vehicle-signal integration required**
Wearable heart-rate / physiological	Heart rate / HRV from wrist sensor plus ML (e.g. Empatica E4)	Above 92% accuracy in user-dependent setting; 30 participants, high-fidelity driving [8]	Low to medium	Windowed; window sets latency	Worn on wrist; motion artefacts; unaffected by light	Health data (physiological)	Best results need per-user training [8]; signal degrades with movement	**Nothing worn by driver; privacy-preserving, on-device facial features**
Commercial OEM DDAW (reference)	Vehicle-signal cues or driver-facing camera, integrated	No public, comparable accuracy. EU mandates DDAW: new types from 6 Jul 2022, all new vehicles from 7 Jul 2024	In vehicle	Yes; production systems	Contactless; camera variants affected by light	Depends on design	Proprietary; internal specifications not published	**Retrofit-friendly and open-stack; designed for Pi/laptop rather than factory integration**

* Cost is a qualitative class, not a measured price. Accuracy values come from different datasets, subjects and label definitions and must not be ranked against each other. Columns 3 and 5 report published experimental results; cost, comfort and privacy are the authors' qualitative assessments.

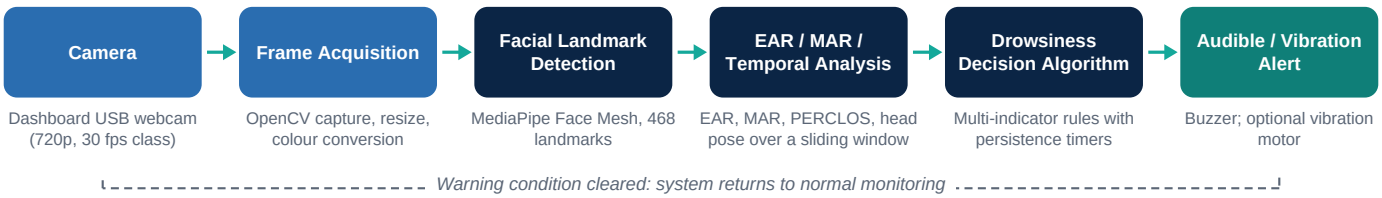
5 Research Gap

Limitation	Observed in existing work	What this project will examine
Hardware cost	EEG and clinical-grade sensors need specialised equipment [6][8]	Commodity webcam plus Raspberry Pi or laptop; cost reported in Section 8
Driver comfort	Electrodes and wearables are contact-based [5][6][8]	Contactless sensing; nothing worn by the driver
Real-time processing	Heavy deep models can strain embedded hardware	Lightweight landmark and rule-based logic; FPS and latency measured on target hardware
Lighting sensitivity	Camera methods degrade in low light and glasses or occlusion [5]	Not solved here: low-light and glasses tests will quantify it; IR camera is future scope
False alarms	Single-cue rules confuse blinks, talking and glances with drowsiness	Multi-indicator fusion, persistence timers and per-driver calibration; false-alarm rate measured
Privacy	Continuous face video is personal data	On-device processing, no frame storage by default, informed consent for test data
Practical deployment	Validation is often simulator-only [6][7][8]	Controlled and recorded-footage tests; deployment limits documented

The project targets these limitations as design goals. Whether it improves on existing systems can only be judged from its own experimental results; no superiority is claimed.

6 Proposed Solution

A dashboard-mounted webcam streams video to a Raspberry Pi or laptop running Python and OpenCV; these are free, well documented and run without a GPU. MediaPipe Face Mesh returns 468 facial landmarks per frame [5]. From these, EAR measures eye openness, MAR measures mouth opening and PERCLOS gives the fraction of a sliding window with eyes closed. Individual cues are noisy, so the decision algorithm fuses sustained eye closure, abnormal blink rate, repeated yawns and optional head nodding, and alerts only when a threshold persists over time, reducing false alarms from normal blinks, talking or brief glances. A buzzer sounds, with an optional vibration motor; monitoring resumes once the condition clears.



7 Implementation Methodology

Phase	Main activities	Deliverable	Indicative time
1. Literature review and hardware selection	Review drowsiness-detection literature and public datasets (e.g. NTHU-DDD); select camera, processor and alert hardware; draft consent and privacy plan	Requirements sheet, bill of materials	Weeks 1–2
2. Camera integration and landmark detection	OpenCV capture pipeline; MediaPipe Face Mesh; test across faces and lighting; record baseline FPS	Live landmark overlay	Weeks 3–4
3. Eye-closure and yawning detection	Compute EAR, MAR, PERCLOS; start from thresholds reported in [5] (EAR 0.2, MAR 0.5, about 0.5 s closure) and calibrate per driver; optional head pose	Indicator module with tunable thresholds	Weeks 5–7
4. Real-time alert integration	Decision algorithm fusing indicators; buzzer and vibration motor via GPIO; reset logic; latency logging	End-to-end prototype	Weeks 8–9
5. Testing, calibration and evaluation	Scenario tests, annotation, metric computation, threshold tuning, error analysis	Performance report, limitations list	Weeks 10–12

Durations are indicative and should be adjusted to the academic calendar.

8 Hardware and Software

Component	Purpose	Approx. cost (INR)
Raspberry Pi 4 (4 GB), or an existing laptop	Runs the vision pipeline and decision logic	Rs. 5,650 (listing: Rs. 5,649 at Robu.in)
Logitech C270 USB webcam (720p, 30 fps)	Driver face capture	Rs. 2,200 (listing: Rs. 2,195)
5 V buzzer	Audible alert	Rs. 40 (Robu.in: Rs. 38 for a 5-piece pack)
Coin vibration motor with driver (optional)	Secondary haptic alert	Rs. 150 (author estimate)
microSD card, power supply, wiring, mount	Storage, power, integration	Rs. 2,000 (allowance, author estimate)
Python, OpenCV, MediaPipe, scikit-learn	Development and optional classifier	Rs. 0 (open source)
Estimated prototype total (Raspberry Pi route)		**About Rs. 10,000 (laptop-only route: Rs. 0–2,200)**

All prices are approximate retailer listings checked in October 2026 (Pi 4 4GB: Robu.in Rs. 5,649; C270: Reliance Digital Rs. 2,195; buzzer: Robu.in) and vary by seller and date. Items marked estimate are budgeting allowances; confirm before purchase.

9 Performance Evaluation

Metric	How it will be measured
Precision	True alerts / all alerts raised
Recall	Annotated drowsiness events detected / all annotated events
F1-score	Harmonic mean of precision and recall
False alarm rate	False alerts per hour of normal-driving footage
Detection latency	Seconds from annotated event onset to alert
Frames per second	Processed frames per second on Pi and on laptop

Test scenarios

- Normal driving
- Prolonged eye closure (requested and simulated)
- Frequent yawning
- Low-light conditions
- Drivers wearing glasses
- Different head orientations

Ethics and safety. Written consent, on-device processing, deletion of raw video after annotation. Testing uses recorded footage, a simulator or a stationary controlled setup; drowsiness is never induced on public roads.

These are planned evaluation metrics, not achieved results. No performance figure is claimed for the prototype.

10 Expected Outcomes

- A working prototype on Raspberry Pi or laptop
- Real-time facial monitoring from a single camera
- Detection of observable fatigue indicators (eye closure, blinking, yawning, optional head pose)
- An immediate audible driver alert, with optional vibration
- Measured precision, recall, F1, false-alarm rate, latency and FPS
- A documented list of deployment limitations

11 Future Scope

- Infrared night-vision camera to reduce lighting dependence
- Edge AI optimisation for lower latency on embedded hardware
- Optional GPS-based emergency notification
- Integration with vehicle safety systems

12 References

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